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Q1

Password validator

```
package model;

/**
 *
 * @author user
 */

public class PasswordValidator {

    public boolean isValid(String password) {

        if (password == null) {
            return false;
        }

        return password.length() >= 8;

    }

}
```

Password validator test

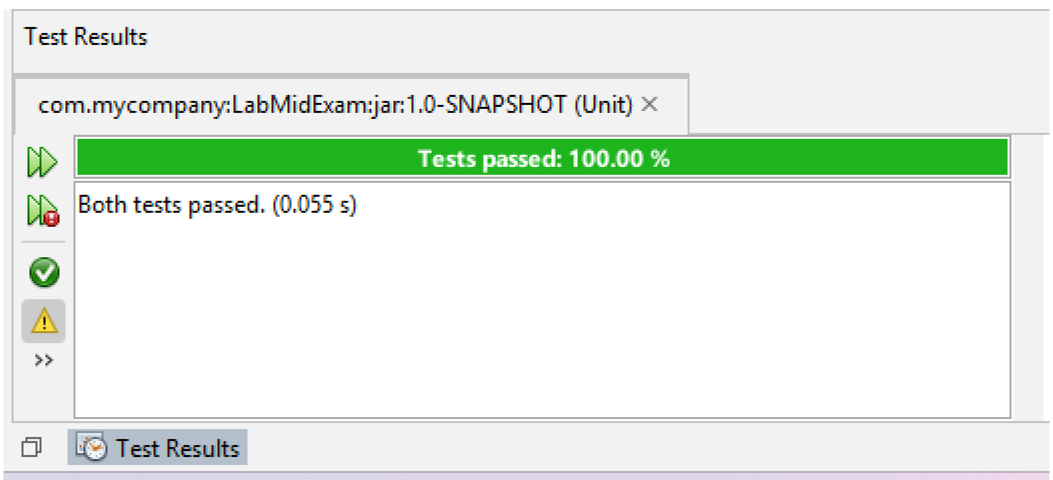
Positive

```java
package test;

/**
 *
 * @author user
 */

import model.PasswordValidator;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class PasswordValidatorTest {

    // Positive Test
    @Test
    void shouldAcceptValidPassword() {

        PasswordValidator validator =
                new PasswordValidator();

        assertTrue(
                validator.isValid("Password123")
        );

    }
    // Negative Test
    @Test
```

Negative

```java
    }
    // Negative Test
    @Test
    void shouldRejectShortPassword() {

        PasswordValidator validator =
                new PasswordValidator();

        assertFalse(
                validator.isValid("abc")
        );

    }

}
```

Output



Q2

Case study : school college pick and drop management system

Use case: assign bus to students

Student

```java
package model;
public class Student {
    private String name;
    private String studentId;
    private Bus bus;
    public Student(
            String name,
            String studentId) {
        this.name = name;
        this.studentId = studentId;

    }
    public String getName() {
        return name;
    }

    public String getStudentId() {
        return studentId;
    }

    public Bus getBus() {
        return bus;
    }

    public void setBus(Bus bus) {
        this.bus = bus;
    }

}
```

Bus

```java
package model;

public class Bus {

    private String busNo;
    private String route;

    public Bus(
            String busNo,
            String route) {

        this.busNo = busNo;
        this.route = route;

    }

    public String getBusNo() {
        return busNo;
    }

    public String getRoute() {
        return route;
    }

}
```

Transport manager

```java
package model;

/**
 *
 * @author user
 */


public class TransportManager {

    public void assignBus(
            Student student,
            Bus bus) {

        student.setBus(bus);

    }

}
```

Transport manager test

```java
package test;
import model.Bus;
import model.Student;
import model.TransportManager;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class TransportManagerTest {
    @Test
    void shouldAssignBusToStudent() {
        TransportManager manager =
                new TransportManager();
        Student student =
                new Student(
                        "Ali",
                        "S001"
                );
        Bus bus =
                new Bus(
                        "BUS-101",
                        "Route A"
                );
        manager.assignBus(
                student,
                bus
        );
        assertEquals(
                bus,
                student.getBus()
        );
    }
}
```

Output